

HISTOLOGY PRACTICE QUESTIONS

COMPREHENSIVE RESPIRATORY SYSTEM STUDY BANK — 50 HIGH-YIELD QUESTIONS

Welcome to the High-Yield Respiratory Histology Practice Bank. This exam-grounded workbook features 50 highly calibrated multiple-choice questions (MCQs) covering the entire microscopic anatomy of the respiratory tract. Every question is mapped directly to core histology curriculum standards and includes a comprehensive explanation to foster active recall and spatial understanding of cell structures, mucosal transitions, and underlying pathological mechanisms. Use these questions for active recall, self-assessment, and clinical shelf examination preparation.

Q1. The respiratory system is divided into conducting and respiratory portions. Which of the following structures represents the final component of the conducting portion?

- A) Respiratory bronchiole
- B) Terminal bronchiole
- C) Alveolar duct
- D) Segmental bronchus

Correct Answer: B

Explanation: The conducting portion serves to deliver, warm, and humidify air, extending from the nasal cavity down to the terminal bronchioles. The respiratory portion, where actual gas exchange occurs, begins at the respiratory bronchioles.

Q2. Epithelial lining down the respiratory tree undergoes a gradual transition in height and complexity. Which of the following correctly describes the epithelial shift from the trachea to the terminal bronchioles?

- A) Tall pseudostratified ciliated columnar to simple cuboidal/columnar
- B) Stratified squamous nonkeratinized to tall pseudostratified ciliated
- C) Simple columnar ciliated to stratified squamous nonkeratinized
- D) Pseudostratified ciliated columnar to simple squamous

Correct Answer: A

Explanation: The typical respiratory epithelium (pseudostratified columnar ciliated with goblet cells) transitions gradually to a shorter ciliated simple columnar epithelium in the larger bronchioles, and eventually simple cuboidal/columnar in the terminal bronchioles.

Q3. Goblet cells are specialized mucus-secreting cells abundant in the upper conducting tract. At what level of the respiratory tree do goblet cells normally disappear?

- A) Primary bronchi
- B) Segmental bronchi
- C) Terminal bronchioles
- D) Respiratory bronchioles

Correct Answer: C

Explanation: Goblet cells are numerous in the trachea but progressively decrease down the bronchial tree. They disappear entirely within the terminal bronchiole, ensuring that mucus is not carried into the respiratory zone where it could block gas exchange.

Q4. The lamina propria of the respiratory mucosa contains Bronchus-Associated Lymphoid Tissue (BALT). Which of the following immunoglobulins is primarily produced by lymphocytes in BALT to prevent pathogens from penetrating the mucosal epithelium?

- A) IgG
- B) IgM
- C) Secretory IgA
- D) IgE

Correct Answer: C

Explanation: Lymphocytes in the BALT (part of mucosal-associated lymphoid tissue, MALT) produce secretory Immunoglobulin A (IgA). Secretory IgA is released onto the mucosal surface where it neutralizes bacteria and viruses, preventing epithelial penetration.

Q5. A researcher is examining the histological layer of a trachea and compares it to a lobar bronchus. Which of the following layers is present in extrapulmonary airways like the trachea but is ABSENT in intrapulmonary bronchi?

- A) Mucosa
- B) Lamina propria
- C) Distinct submucosa
- D) Adventitia

Correct Answer: C

Explanation: A distinct submucosa layer is present in extrapulmonary airways only. In intrapulmonary bronchi, there is no distinct submucosa because glands and lymphatic tissue lie directly between the cartilage plates and in the adventitia.

Q6. Autonomic nerves regulate blood flow in the nasal cavity. In some regions of the nasal mucosa, specialized thin-walled veins with wide lumina act similarly to erectile tissue. What is the correct term for these vascular structures?

- A) Cavernous sinusoids
- B) Swell or cavernous bodies
- C) Vasa vasorum
- D) Lymphatic sinusoids

Correct Answer: B

Explanation: Swell or cavernous bodies are specialized veins resembling erectile tissue in the nasal mucosa. They engorge with blood periodically and alternatively on each side of the nasal cavity, closing that side and giving the mucosa time to recover from drying.

Q7. The typical respiratory epithelium appears stratified under light microscopy, hence the name 'pseudostratified'. What is the true morphological arrangement of these cells relative to the basement membrane?

- A) All cells reach the apical lumen but some do not touch the basement membrane
- B) Cells are stacked in multiple distinct layers, with only the basal layer touching the membrane
- C) All cells rest on the basement membrane but not all reach the apical surface
- D) Only ciliated cells contact the basement membrane

Correct Answer: C

Explanation: In pseudostratified columnar ciliated epithelium, all cells sit on the basement membrane, but because they have different shapes and heights, not all reach the luminal surface. This creates multiple rows of nuclei, mimicking a stratified tissue.

Q8. Three main cell types make up over 90% of the respiratory epithelial cell population under light microscopy. Which of the following is NOT one of these three abundant cell types?

- A) Ciliated columnar cells
- B) Goblet cells
- C) Basal cells
- D) Brush cells

Correct Answer: D

Explanation: The three cells visible via light microscopy making up over 90% of the cells are: (1) ciliated columnar, (2) goblet, and (3) basal cells. Brush and granule cells constitute less than 10% of the population and require electron microscopy or special staining for clear visualization.

Q9. Basal cells of the respiratory epithelium are small, rounded cells located along the basement membrane. What is their primary functional significance?

- A) Act as stem cells that divide by mitosis to regenerate other epithelial cell types
- B) Transduce general sensations from the airway lumen
- C) Secrete catecholamines to regulate smooth muscle tone
- D) Form tight junctions to prevent paracellular fluid movement

Correct Answer: A

Explanation: Basal cells are undifferentiated progenitor/stem cells. They divide by mitosis and differentiate to replace ciliated, goblet, and other cells when they are damaged or naturally turn over.

Q10. An electron micrograph of trachea epithelium reveals specialized columnar cells carrying apical microvilli and containing cytoplasmic glycogen. These cells are specialized for the transduction of general sensation. What are these cells?

- A) Granule cells
- B) Basal cells
- C) Brush cells
- D) Clara cells

Correct Answer: C

Explanation: Brush cells are columnar cells characterized by numerous apical microvilli and cytoplasmic glycogen. They are specialized for general sensation transduction and may also represent immature columnar or degranulated goblet cells.

Q11. Granule cells are neuroendocrine cells of typical respiratory epithelium. Based on the function of their cytoplasmic granules, these cells belong to which of the following categories?

- A) Catecholamine-secreting (neurosecretory) and protein-hormone secreting (enteroendocrine-like)
- B) Secretory IgA-producing and surfactant-secreting
- C) Mucus-secreting and OBP-secreting
- D) Lysosomal-producing and elastase-secreting

Correct Answer: A

Explanation: Granule cells contain secretory granules of two types: catecholamine-secreting cells (neurosecretory) and protein hormone-secreting cells (enteroendocrine-like cells), which secrete amines and amine precursors.

Q12. The pulmonary vasculature operates under significantly lower hydrostatic pressures than systemic circulation. What is the typical systolic and diastolic pressure within the functional pulmonary circuit?

- A) 120 mmHg systolic, 80 mmHg diastolic
- B) 80 mmHg systolic, 40 mmHg diastolic
- C) 25 mmHg systolic, 5 mmHg diastolic
- D) 10 mmHg systolic, 2 mmHg diastolic

Correct Answer: C

Explanation: Because the pulmonary circuit is a low-pressure system (typically 25 mmHg systolic and 5 mmHg diastolic), branches of the pulmonary artery have relatively thin walls compared to systemic arteries of similar caliber.

Q13. A biopsy shows a branch of the pulmonary artery (PA) traveling within the lung. In which of the following histological structures will this artery branch be located?

- A) Within the inner mucosal layer of the alveoli
- B) Within the adventitia of the bronchi and bronchioles
- C) Within the submucosal gland duct system of the trachea
- D) Running free inside the pleural space

Correct Answer: B

Explanation: Within the lung, the pulmonary artery branches and closely accompanies the bronchial tree. Its branches are surrounded by the adventitia of the bronchi and bronchioles, before forming capillaries at the alveolar level.

Q14. The conchae (turbinates) project into the nasal cavity. How do they increase the warming and filtration efficiency of inspired air?

- A) By accelerating the flow rate to match body temperature
- B) By increasing surface area and breaking the air stream into eddies (turbulent precipitation)
- C) By secreting thick surfactant that trap particles
- D) By paralyzing incoming bacteria with high salt concentrations

Correct Answer: B

Explanation: The conchae increase the mucosal surface area and create eddies in the air stream (turbulent precipitation). This causes suspended dust and particles to be thrown out of the stream and adhere to the mucus-covered nasal wall, where they are moved to the pharynx by cilia and swallowed.

Q15. The paranasal sinuses are bilateral air-filled cavities in the skull bones. How do their histological linings compare to the main nasal cavities?

- A) Lined with nonkeratinized stratified squamous epithelium
- B) Completely lack ciliated cells and contain massive serous glands
- C) Lined with a thinner respiratory mucosa, thinner epithelium, fewer goblet cells, and thin lamina propria
- D) Contain a thick submucosa packed with cavernous erectile tissue

Correct Answer: C

Explanation: Paranasal sinuses are lined with a thinner respiratory mucosa, featuring a thinner respiratory epithelium, fewer goblet cells, and a very thin lamina propria that contains only a few small mucous glands and is continuous with the bone's periosteum.

Q16. The olfactory mucosa is the receptor organ for smell. What pigment is responsible for the characteristic yellowish-brown color of this mucosa?

- A) Melanin
- B) Bilirubin
- C) Hemosiderin
- D) Lipofuscin

Correct Answer: D

Explanation: Lipofuscin granules are highly prevalent within both the sustentacular cells of the olfactory epithelium and the secretory cells of Bowman's glands. Together, they give the olfactory mucosa its natural yellow-brown coloration.

Q17. Olfactory sustentacular cells play a metabolic and physical role supporting the sensory receptor cells. Which of the following junctions connect these supporting cells to olfactory receptor cells?

- A) Adhering junctions
- B) Gap junctions
- C) Tight junctions
- D) Desmosomes only

Correct Answer: A

Explanation: Adhering junctions are present between the sustentacular cells and the olfactory receptor (sensory) cells. However, gap and tight junctions are completely absent, allowing the olfactory receptor cells to remain structurally separate.

Q18. What type of glands are Bowman's glands, located in the olfactory lamina propria?

- A) Pure mucous tubulo-alveolar glands
- B) Mixed seromucous alveolar glands
- C) Pure serous branched tubulo-alveolar glands
- D) Sebaceous branched glands

Correct Answer: C

Explanation: Bowman's glands (olfactory glands) are pure serous, branched tubulo-alveolar glands. They produce a thin, watery secretion that dissolves odorous substances and washes the cilia, preparing receptors to receive new stimuli.

Q19. The auditory (Eustachian) tube permits equilibration of air pressure in the middle ear. What are the histological and structural characteristics of its wall?

- A) Lined by simple squamous epithelium and supported by elastic cartilage
- B) Lined by typical respiratory epithelium and supported by elastic cartilage
- C) Lined by stratified squamous epithelium and supported by hyaline cartilage
- D) Lined by stratified cuboidal epithelium and supported by skeletal muscle

Correct Answer: B

Explanation: The Eustachian tubes are lined by typical respiratory epithelium (ciliated pseudostratified columnar with goblet cells) and receive structural support from elastic cartilage within their walls.

Q20. The pharyngeal wall is composed of three principal layers. Which of the following correctly pairs the region of the pharynx with its lining epithelium?

- A) Nasopharynx - Simple cuboidal epithelium
- B) Nasopharynx - Ciliated pseudostratified columnar with goblet cells
- C) Oropharynx - Tall pseudostratified ciliated columnar
- D) Laryngopharynx - Simple columnar epithelium

Correct Answer: B

Explanation: The lining epithelium of the pharynx is region-specific: ciliated pseudostratified columnar with goblet cells in the nasopharynx, and nonkeratinized stratified squamous epithelium in the oropharynx and laryngopharynx.

Q21. During swallowing, the glottis is closed off to prevent food entry. Which of the following describes the core structure of the epiglottis that facilitates its flexible movement?

- A) Hyaline cartilage plate
- B) Core of elastic cartilage
- C) Skeletal vocalis muscle bundle
- D) Concentric rings of fibrocartilage

Correct Answer: B

Explanation: The epiglottis consists of a central core of elastic cartilage, which provides the flexibility required to cover and close the laryngeal glottis opening during swallowing.

Q22. The epiglottis has distinct anterior (lingual) and posterior (laryngeal) mucosal surfaces. Which of the following represents the correct epithelial transition on its laryngeal surface?

- A) Continuous nonkeratinized stratified squamous throughout the posterior surface
- B) Nonkeratinized stratified squamous at the apical portion, transitioning to typical respiratory epithelium
- C) Simple columnar ciliated at the apical portion, transitioning to simple squamous
- D) Simple squamous mesothelium throughout

Correct Answer: B

Explanation: The entire anterior (lingual) surface and the apical portion of the posterior (laryngeal) surface of the epiglottis are covered by nonkeratinized stratified squamous epithelium. At a variable point on the laryngeal surface, it transitions to typical respiratory epithelium.

Q23. The larynx contains two pairs of mucosal folds. Which of the following correctly describes the False Vocal Cords (vestibular folds)?

- A) Covered by stratified squamous epithelium and containing no glands
- B) Covered by typical respiratory epithelium and containing seromucous glands and areolar tissue
- C) Devoid of lymphoid tissue and containing a thick elastic ligament
- D) Lined by simple columnar epithelium and containing skeletal muscle

Correct Answer: B

Explanation: The False Vocal Cords (vestibular folds) are the upper paired folds, covered by typical respiratory epithelium, and contain seromucous glands and areolar tissue with MALT (often containing lymphoid nodules).

Q24. Skeletal muscles in the larynx allow for precise adjustments during vocalization. Which laryngeal fold contains the vocalis muscle and the vocal ligament?

- A) False vocal cord (vestibular fold)
- B) True vocal cord (vocal fold)
- C) Epiglottis
- D) Ventricle

Correct Answer: B

Explanation: The True Vocal Cords (vocal folds) contain the vocalis muscle (striated skeletal muscle) and the vocal ligament (a thick band of elastic fibers) covered by protective nonkeratinized stratified squamous epithelium.

Q25. The trachea contains U- or C-shaped hyaline cartilage segments. What smooth muscle bridges the free posterior ends of these cartilage rings?

- A) Vocalis muscle
- B) Trachealis muscle
- C) Pharyngeal constrictor
- D) Muscularis mucosae

Correct Answer: B

Explanation: The 16-20 U- or C-shaped pieces of tracheal hyaline cartilage are joined posteriorly at their free ends by bands of smooth muscle called the trachealis muscle.

Q26. The primary bronchi enter the lungs at the hilum. How are the secondary (lobar) bronchi distributed between the right and left lungs?

- A) Three in the right lung, three in the left lung
- B) Two in the right lung, three in the left lung
- C) Three in the right lung, two in the left lung
- D) One in the right lung, two in the left lung

Correct Answer: C

Explanation: At the hilum, primary bronchi give rise to three secondary (lobar) bronchi in the right lung (supplying three lobes) and two in the left lung (supplying two lobes).

Q27. A thoracic surgeon plans to resect a diseased bronchopulmonary segment. What structural and anatomical features of this segment facilitate specific surgical resection?

- A) It is supplied by a primary bronchus and has its own separate pleural cavity
- B) It represents 10-12% of the lung and has its own connective tissue capsule and blood supply
- C) It lacks any blood supply and is ventilated purely by collateral pores
- D) It is defined strictly by the absence of cartilage plates

Correct Answer: B

Explanation: A bronchopulmonary segment (10-12% of lung) is supplied by a tertiary (segmental) bronchus. Because it is anatomically isolated with its own connective tissue capsule, air supply, and blood supply, specific diseased segments can be safely resected surgically.

Q28. Anatomical septa delineate the pyramid-shaped pulmonary lobules. In which of the following groups are these connective tissue septa most completely developed and visible?

- A) Adults
- B) Elderly smokers
- C) Fetuses
- D) Emphysema patients

Correct Answer: C

Explanation: Each pulmonary lobule is delineated by a thin connective tissue septum. This septum is best seen and completely delineated in the fetus, whereas in adults, the septa are frequently incomplete, resulting in poor lobular delineation.

Q29. Histological sections of intrapulmonary secondary bronchi exhibit a folded mucosa. Which of the following layers is directly responsible for this characteristic folded appearance?

- A) A thick adventitial elastic layer
- B) Skeletal muscle fibers of the trachealis
- C) A complete layer of crisscrossing spirally arranged smooth muscle (muscularis mucosae)
- D) Swell bodies in the lamina propria

Correct Answer: C

Explanation: The intrapulmonary secondary bronchus mucosa is folded in histologic sections. This is caused by the contraction of a complete layer of smooth muscle fibers (muscularis mucosae) that are spirally arranged and crisscross beneath the lamina propria.

Q30. How does the distribution of hyaline cartilage change in the wall of an intrapulmonary bronchus compared to the trachea?

- A) Cartilage is completely absent in the intrapulmonary bronchus
- B) Cartilage exists as single, continuous semilunar plates
- C) Cartilage is broken up into discontinuous, interconnected hyaline cartilage plates
- D) Cartilage is replaced entirely by elastic cartilage rings

Correct Answer: C

Explanation: Unlike the continuous C-shaped rings in the trachea and primary bronchi, cartilage in the intrapulmonary bronchus wall is broken up into discontinuous, interconnected hyaline cartilage plates.

Q31. Bronchioles are intralobular airways with an outer diameter of 1mm or less. Which of the following features is a key defining landmark of a bronchiole?

- A) Presence of simple squamous lining
- B) Complete lack of cartilage support and seromucous glands
- C) Absence of a smooth muscle layer
- D) Abundance of goblet cells

Correct Answer: B

Explanation: Bronchioles are structurally defined as intralobular airways that completely lack cartilage plates and seromucous glands within their walls.

Q32. The lack of cartilage support in bronchioles represents a significant clinical hazard. In which of the following clinical conditions do the bronchioles excessively constrict, closing the airway lumen?

- A) Emphysema
- B) Bronchial Asthma
- C) Immotile Cilia Syndrome
- D) Hyaline Membrane Disease

Correct Answer: B

Explanation: Because bronchioles lack cartilage, they are prone to collapse if bronchiolar smooth muscle tone increases. During asthma (an allergic response to irritants), bronchoconstriction leads to wheezing and difficulty in expiration.

Q33. Bronchiolar smooth muscle is under dual autonomic nervous system control. Which of the following matches the neural stimulation to its physiological effect on the airway diameter?

- A) Vagus nerve (parasympathetic) - Bronchodilation
- B) Sympathetic stimulation - Bronchoconstriction
- C) Vagus nerve (parasympathetic) - Bronchoconstriction
- D) Epinephrine administration - Bronchoconstriction

Correct Answer: C

Explanation: Stimulation of the vagus nerve (parasympathetic) causes contraction of bronchiolar smooth muscle, decreasing airway diameter (bronchoconstriction). Sympathetic stimulation (and sympathomimetic drugs like epinephrine) produces the opposite effect (bronchodilation).

Q34. A terminal bronchiole represents the smallest purely conducting airway. What epithelium lines the mucosa of a terminal bronchiole?

- A) Tall pseudostratified ciliated columnar with goblet cells
- B) Ciliated simple columnar or cuboidal epithelium without goblet cells
- C) Simple squamous epithelium
- D) Nonkeratinized stratified squamous

Correct Answer: B

Explanation: The mucosa of terminal bronchioles is characterized by a simple ciliated columnar or cuboidal epithelium that is completely free of goblet cells.

Q35. Terminal bronchioles contain dome-shaped, non-ciliated secretory cells interspersed in the ciliated epithelium. What is the correct current name for these cells?

- A) Dust cells
- B) Goblet cells
- C) Club cells
- D) Granule cells

Correct Answer: C

Explanation: These non-ciliated secretory cells are currently called Club cells. They were formerly known as Clara cells.

Q36. On January 1, 2013, the name 'Clara cell' was officially replaced by 'Club cell'. What was the primary reason for this nomenclature change?

- A) The cells were discovered to belong to a lymphoid lineage
- B) Max Clara used lung tissue from executed victims of the Nazi regime
- C) A new, more specific receptor was discovered on their surface
- D) To avoid confusion with typical goblet cells

Correct Answer: B

Explanation: Dr. Max Clara first described these cells in 1937 using lung tissue obtained from victims executed by the Nazi's Third Reich. Due to the ethical controversy surrounding this, the scientific community officially adopted the name 'Club cell'.

Q37. Club cells produce a 16-kilodalton secretory protein (CC16). What is the physiological role of CC16 in the terminal bronchiole fluid layer?

- A) Acts as a surfactant-like product that alters surface tension
- B) Phagocytoses dust and complexed iron
- C) Directly triggers bronchoconstriction in response to histamine
- D) Combines with secretory IgA to kill bacteria

Correct Answer: A

Explanation: CC16 is a major phospholipid/lipoprotein-rich secretory product. It acts as a surfactant-like agent that alters the surface tension of the fluid layer covering the bronchiolar surface, keeping it moist and preventing collapse.

Q38. CC16 is utilized clinically as a biomarker. How do the levels of CC16 in airway fluid and serum change during lung injury?

- A) CC16 increases in airway fluid and decreases in serum
- B) CC16 decreases in airway fluid and increases in serum
- C) CC16 decreases in both airway fluid and serum
- D) CC16 remains unchanged in airway fluid but disappears from serum

Correct Answer: B

Explanation: During lung injury (such as in COPD or asthma), damage to Club cells decreases CC16 secretion into bronchiolar fluid. However, due to leakage across the damaged air-blood barrier, serum levels of CC16 may increase.

Q39. Why are respiratory bronchioles termed 'respiratory' rather than 'conducting'?

- A) They are lined by simple squamous epithelium
- B) They are surrounded by continuous hyaline cartilage plates
- C) Their walls are interrupted at intervals by the openings of individual alveoli where gas exchange occurs
- D) They contain ciliated cells only

Correct Answer: C

Explanation: Respiratory bronchioles are named because they mark the start of the respiratory zone. Their simple cuboidal ciliated wall is interrupted at irregular intervals by the openings of saclike individual alveoli, allowing gas exchange to take place.

Q40. At the junction where a respiratory bronchiole meets an alveolus, what epithelial transition occurs?

- A) Simple cuboidal transitions directly to the simple squamous epithelium of Type I pneumocytes
- B) Simple cuboidal transitions to pseudostratified ciliated columnar
- C) Simple columnar transitions to stratified squamous
- D) Simple squamous transitions to simple columnar

Correct Answer: A

Explanation: At the junction between the respiratory bronchiole and the alveoli, the simple cuboidal epithelial cells continue as the thin simple squamous epithelium (specifically Type I pneumocytes) of the alveolus wall.

Q41. Each adult lung contains about 150 to 250 million alveoli. What is their combined internal surface area available for gas exchange?

- A) Approximately 10 square meters
- B) Approximately 75 to 140 square meters
- C) Over 500 square meters
- D) Less than 5 square meters

Correct Answer: B

Explanation: The combined surface area of all alveoli in an adult is calculated to be between 75 m² and 140 m² (roughly equivalent to the size of a tennis court), providing a vast interface for gas exchange.

Q42. An interalveolar septum separates adjacent alveoli. Which of the following is NOT a cellular or fibrous component of this septum?

- A) Elastic and reticular fibers
- B) Non-fenestrated continuous blood capillaries
- C) Skeletal vocalis muscle fibers
- D) Contractile septal cells containing actin and myosin

Correct Answer: C

Explanation: The interalveolar septum (lung interstitium) consists of elastic and reticular fibers, fibroblasts, macrophages, continuous capillaries, and contractile septal cells containing actin and myosin (which contract in response to hypoxia). Skeletal muscle is absent (found only in the pharynx/larynx).

Q43. Which of the following describes the function of the alveolar pores of Kohn in the interalveolar septum?

- A) They secrete surfactant to reduce surface tension
- B) They equalize pressure inside the alveoli and allow collateral air circulation
- C) They prevent the spread of lung infections to adjacent alveoli
- D) They anchor the lung to the visceral pleura

Correct Answer: B

Explanation: Alveolar pores of Kohn connect adjacent alveoli, allowing equalization of air pressure within a lobule and permitting collateral air circulation if a local bronchiole is obstructed. However, they may also facilitate the spread of infections.

Q44. Type I pneumocytes line almost the entire alveolar surface. What is their population percentage versus surface area coverage?

- A) 8% of the cell population, covering 95% of the surface area
- B) 95% of the cell population, covering 8% of the surface area
- C) 16% of the cell population, covering 5% of the surface area
- D) 40% of the cell population, covering 60% of the surface area

Correct Answer: A

Explanation: Because Type I pneumocytes are highly flattened squamous cells, they form only 8% of the total alveolar cell population but cover 95% of the alveolar surface area to facilitate optimal gas diffusion.

Q45. Type II pneumocytes are large, rounded cuboidal cells. What is their population percentage versus surface area coverage?

- A) 8% of the cell population, covering 95% of the surface area
- B) 16% of the cell population, covering 5% of the surface area
- C) 30% of the cell population, covering 30% of the surface area
- D) 60% of the cell population, covering 95% of the surface area

Correct Answer: B

Explanation: Type II pneumocytes (septal cells) are cuboidal and make up 16% of the total septum cell population (representing 60% of the pneumocytes), but they cover only 5% of the alveolar surface area.

Q46. Under transmission electron microscopy, Type II pneumocytes exhibit highly unique, membrane-bound secretory vesicles containing parallel or concentric lamellae. What are these organelles?

- A) Multivesicular bodies
- B) Lamellar bodies
- C) Melanosomes
- D) Mucigen granules

Correct Answer: B

Explanation: Lamellar bodies are 1-2 μm diameter secretory vesicles limited by a unit membrane (phospholipid monolayer) containing concentric parallel lamellae. They store newly synthesized pulmonary surfactant.

Q47. Surfactant secreted by Type II pneumocytes is critical for lung function. What is its primary physiological role?

- A) Warm inspired air via countercurrent heat exchange
- B) Inhibit elastase secretion by neutrophils
- C) Decrease alveolar surface tension forces to prevent collapse
- D) Facilitate the direct transport of oxygen molecules into erythrocytes

Correct Answer: C

Explanation: Surfactant decreases alveolar surface tension forces to a minimal level, preventing alveoli from collapsing during expiration. Its presence is vital for newborns to successfully obtain their first breath of air.

Q48. Inhalation of toxic gases or similar materials can destroy the highly sensitive Type I pneumocytes lining the alveoli. How are these squamous cells replaced during lung regeneration?

- A) Endothelial cells of continuous capillaries transdifferentiate
- B) Remaining Type II cells divide and differentiate into both Type I and Type II cells
- C) Macrophages transdifferentiate into squamous epithelial cells
- D) Basal cells from the trachea migrate distally

Correct Answer: B

Explanation: Type II pneumocytes act as mitotic stem cells for the alveolar lining. If Type I cells are killed, the remaining Type II cells undergo mitotic division, and their progeny differentiate to renew and restore both alveolar cell types.

Q49. Alveolar macrophages (dust cells) are highly active in the lungs. What is their origin, and why do they often appear dark under light microscopy?

- A) Derived from fibroblasts; appear dark due to lipofuscin granules
- B) Derived from monocytes; appear dark due to phagocytosed carbon/dust and hemosiderin
- C) Derived from basal cells; appear dark due to glycogen accumulations
- D) Derived from Club cells; appear dark due to dense lamellar body aggregates

Correct Answer: B

Explanation: Alveolar macrophages are phagocytic cells derived from blood monocytes. They appear darker in lung sections due to accumulated carbon/dust from inhaled air and hemosiderin (complexed iron) from phagocytosing extravasated erythrocytes.

Q50. The Blood-Air Barrier (pulmonary barrier) separates capillary blood from alveolar air. At its thinnest point, this barrier consists of which of the following components?

- A) Type I pneumocyte cytoplasm, fused basal laminae, and capillary endothelial cytoplasm
- B) Type II pneumocyte cytoplasm, thin interstitium, and fenestrated endothelium
- C) Mesothelial cell cytoplasm, fused basal laminae, and Type I pneumocyte cytoplasm
- D) Surfactant, connective tissue septum, and capillary basal lamina

Correct Answer: A

Explanation: At its thinnest and most efficient points (0.1 to 1.5 μm thick), the blood-air barrier consists of: (1) cytoplasm of the Type I pneumocyte, (2) the fused basal laminae of the alveolar epithelium and capillary endothelium, and (3) cytoplasm of the capillary endothelial cell.